



RoboCup@Home

Rules & Regulations

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This is the official rulebook of the RoboCup@Home competition 2025. The rulebook has been written by the 2025 RoboCup@Home Technical Committee.

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Chapter 1

Introduction

1.1 RoboCup

RoboCup is an international joint project to promote AI, robotics, and related fields. It is an attempt to foster AI and intelligent-robotics research by providing standard problems where a wide range of technologies can be integrated and examined. More information can be found at http://www.robocup.org/.

1.2 RoboCup@Home

The RoboCup@Home league aims to develop service and assistive robot technology with high relevance for future personal domestic applications. It is the largest international annual competition for autonomous service robots and is part of the RoboCup initiative. A set of benchmark tests is used to evaluate the abilities and performance of different robots in a realistic, non-standardized home environment setting. The focus is on, but is not limited to, the following domains: human-robot interaction and cooperation, navigation and mapping in dynamic environments, computer vision and object recognition under natural light conditions, object manipulation, adaptive behaviors, behavior integration, ambient intelligence, standardization and system integration. The competition is co-located with the RoboCup symposium.

1.3 Organization

@Home is organized into three subcommittees. The current members of the committees are listed at https://athome.robocup.org/committees/.

1.3.1 Executive Committee — rc-home-ec@lists.robocup.org

The *executive committee* (EC) consists of members of the board of trustees, and representatives of each activity area, and supervises both the TC and OC. The committee is responsible for the long-term planning of the league and cast the final vote in all decisions within the competition, including those pertaining to any of the committees and any other affair that requires a qualified vote. Additionally, the EC must act when any of the committees fail, as it responds to the Trustee board and directs the league accordingly.

1.3.2 Technical Committee — rc-home-tc@lists.robocup.org

The *technical committee* (TC) is responsible for the rules of the league; its main focus is writing the rulebook and refereeing. The members of the EC are always members of the TC as well.

1.3.3 Organizing Committee — rc-home-oc@lists.robocup.org

The *organizing committee* (OC) is responsible for the organization of the competition, namely it creates the schedule and provides information about the scenarios. The *Local Organizing Committee* (LOC), on the other hand, is responsible for the set up and organization of the competition venue.

1.4 Infrastructure

RoboCup@Home Mailing List The official @Home mailing list can be found at robocup-athome@lists.robocup.org. You can subscribe to the mailing list at: https://lists.robocup.org/mm/lists/robocup-athome.lists.robocup.org/

RoboCup@Home Web Page The official @Home website that also hosts this rulebook can be found at https://athome.robocup.org/

RoboCup@Home Rulebook Repository The official @Home *Rulebook Repository* is where rules are publicly discussed before applying changes. The entire @Home community is welcome and encouraged to actively participate in creating and discussing the rules. The *Rulebook Repository* is hosted at https://github.com/RoboCupAtHome/RuleBook/

RoboCup@Home Telegram Group The official @Home *Telegram Group* is a communication channel for the @Home community where rules are discussed, announcements are made, and questions are answered. Beyond supporting the technical aspects of the competition, the group is a meeting point to stay in contact with the community, foster knowledge exchange, and strengthen relationships. The *Telegram Group* can be reached at https://t.me/RoboCupAtHome

RoboCup@Home Wiki The official @Home *Wiki* is meant to be a central place to collect information on all topics related to the @Home league. The wiki was set up to simplify and unify the exchange of relevant information; this includes, but is not limited to, hardware, software, media, data, and more. The *Wiki* can be reached at https://github.com/RoboCupAtHome/AtHomeCommunityWiki/wiki

1.5 Leagues

@Home is divided into two Leagues. One of these grants complete freedom to all competitors with respect to the robot used, while in the other all competitors use the same robot. The official leagues and their names are:

- *Open Platform League* (OPL)

- *Domestic Standard Platform League* (DSPL)

Each league focuses on a different aspect of service robotics by targeting specific abilities.



Figure 1.1: Toyota HSR

1.5.1 Domestic Standard Platform League (DSPL)

The main goal of the DSPL is to assist humans in a domestic environment, paying special attention to elderly people and people suffering from illness or disability. As a consequence, the DSPL focuses on *Ambient Intelligence*, *Computer Vision*, *Object Manipulation*, safe indoor *Navigation* and *Mapping*, and *Task Planning*. The robot used in the DSPL is the *Toyota HSR*, shown in Figure 1.1.

1.5.2 Open Platform League (OPL)

The OPL has had the same modus operandi since the foundation of @Home. With no hardware constrains, OPL is the league for teams who want to test their own robot designs and configurations, as well as for old at-homers. In this league, robots are tested to their limits without having in mind any concrete design restriction, although the scope is similar to the DSPL.

1.6 Competition

The competition consists of two *Stages* and a *Final*. Each stage consists of a series of *Tests* that are being held in a daily life environment. The best teams from *Stage I* advance to *Stage II*, which consists of more difficult tests. The competition ends with the *Final*, where only the two highest-ranked teams of each league compete to select the winner.

1.7 Awards

The @Home league features the *awards* described below. Note that all awards need to be approved by the *RoboCup Federation* (RCF); based on a decision by the RCF, some of them may not be given.

1.7.1 Winner of the Competition

For each league, there will be 1st, 2nd, and 3rd place award trophies (or first and second place only if the number of teams in a league is eight or less).

1.7.2 Best Human-Robot Interface Award

To honor outstanding human-robot interfaces developed for interacting with robots in @Home, a special *Best Human-Robot Interface Award* may be given to one of the participating teams. Special attention is paid to making the interface open and available to the @Home community.

The @Home EC members nominate a set of candidates for the award and the TC elects the winner. A TC member whose team is among the nominees is not allowed to vote. There is no *Best Human-Robot Interface Award* in case the EC decides that there is no outstanding interface, and thus no nominees.

1.7.3 Best Poster Award

To foster scientific knowledge exchange and reward the teams' efforts to present their research contributions, all scientific posters of each league are evaluated and have the chance of receiving the *Best DSPL Poster Award*or the *Best OPL Poster Award*respectively.

Candidate posters must present innovative and state-of-the-art research within a field with a direct application to @Home, and demonstrate successful and clear results in an easy-to-understand way. In addition to being attractive and well-rated in the *Poster Session* (see Section 4.3), the described research must have impact in the team's performance during the competition.

The @Home EC members nominate a set of candidates for the award and the TC elects the winner. A TC member whose team is among the nominees is not allowed to vote.

1.7.4 Open Challenge Award

To encourage teams to present their research to the other teams in the competition off-hours, @Home grants the *Best Open Challenge Award* to the best open demonstration presented during the competition. This award is granted only if a team has demonstrated innovative research that is related to the global objectives of @Home; thus, the award is not necessarily granted.

The @Home TC members, with a recommendation from the team leaders, nominate a set of candidates for the award; the EC decides if the award should be granted and elects the winner. A TC member is not allowed to nominate their own team without a recommendation from the team leaders.

1.7.5 Skill Certificates

The @Home league features certificates for best demonstrated skills in *Navigation*, *Manipulation*, *Person Recognition*, and *Natural Language Processing*. A team is given the certificate if it scores at least 75% of the attainable points for that skill. This is counted over all tests and challenges, so, for example, if a robot scores manipulation points during the *Help-me-Carry* test to open the door, that will count for the *Manipulation* certificate. Note that the certificate will only be handed out if the team is *not* the overall winner of the competition.

1.7.6 Open-source software award

Since Nagoya 2017, RoboCup@Home awards the best contribution to the community by means of an open source software solution. To be eligible for the award, the software must be easy

to read, have proper documentation, follow standard design patterns, be actively maintained, and meet the IEEE software engineering metrics of scalability, portability, maintainability, fault tolerance, and robustness. In addition, the open sourced software must be made available as a framework-independent standalone library so it can be reused with any software architecture.

Candidates must send their application to the TC at least one month before the competition by means of a short paper (maximum 4 pages), following the same format used for the *Team Description Paper* (TDP) (see Section 3.1.2), including a brief explanation of the approach, comparison with state-of-the-art techniques, statement of the used metrics and software design patterns, and the name of the teams and other collaborators that are also using the software being described.

The @Home TC members nominate a set of candidates for the award and the EC elects the winner. An EC/TC member whose team is among the nominees is not allowed to vote.

Chapter 2

Concepts Behind the Competition

A set of conceptual key criteria builds the basis for the RoboCup@Home competition. These criteria are to be understood as a common agreement on the general concept of the competition. The concrete rules are listed in the @Home Rules & Regulations.

2.1 Lean Set of Rules

To allow for different, general, and transmissible approaches in the @Home competition, the rule set should be as lean as possible. Nonetheless, to avoid rule discussions during the competition itself, it should also be concrete enough to leave no room for diverse interpretations. If, during a competition, there are any discrepancies or multiple interpretations, a decision will be made by the TC and the referees on site.

Note: Once the test scoresheet has been signed or the scores has been published, the TC decision is irrevocable.

2.2 Autonomy & Mobility

The aim of @Home is to foster mobile autonomous service robotics and natural human-robot interaction. Thus, all robots participating in the RoboCup@Home competition must be *mobile* and *autonomous*, which means that humans are not allowed to directly (remotely) control the robot (this also includes verbally remotely controlling the robot).

2.3 Aiming for Applications

To foster the advance in technology and to keep the competition interesting, the scenario and the tests will steadily increase in complexity. While necessary individual abilities are still being tested in the competition, tests will focus more and more on real applications with a rising level of complexity and uncertainty. Useful, robust, general, cost effective, and applicable solutions are rewarded in @Home.

2.4 Social Relevance

The competition and the included tests should produce socially relevant results, as the aim is to convince the public about the usefulness of autonomous robotic applications. This should

be done by showing applications where robots directly help or assist humans in everyday life situations. Examples of such applications are: a personal robot assistant, a guide robot for the blind, robot care for elderly people, and so forth. Such socially relevant results are rewarded in @Home.

2.5 Scientific Value

@Home should not only show what can be put into practice today, but should also present new approaches, even if they are not yet fully applicable or if they demand a very special configuration or setup. Therefore, a high scientific value of an approach is rewarded.

2.6 Time Constraints

To allow for many participating teams and tests as well as to foster a simple setup procedures, the setup time and the time for performing the tests is very limited.

2.7 No Standardized Scenario

The scenario for the competition should be simple but effective, available world-wide, and at low cost. As uncertainty is part of the concept, no standard scenario will be provided in @Home. One can expect that the scenario will look typical for the country where the competition is hosted.

The scenario is something that people encounter in daily life; this can be a domestic environment, such as a living room and a kitchen, but also an office space, a supermarket, a restaurant, etc. The scenario should change from year to year, as long as the desired tests can still be executed. Furthermore, tests may take place outside of the scenario, that is, in a previously unknown environment, such as a public space nearby.

2.8 Attractiveness

The competition should be attractive for the audience and the public; thus, high attractiveness and originality of an approach will be rewarded.

2.9 Community

While they have to compete against each other during the competition, the members of the @Home league are expected to cooperate and exchange knowledge to advance technology together. The *RoboCup@Home mailing list* as well as the *Rulebook Repository* can be used to get in touch with other teams and to discuss league-specific issues such as rule changes, proposals for new tests, etc. In addition, every team is expected to share relevant technical, scientific (and team-related) information in a Team Description Paper and on the team's website.

Finally, all teams are invited to submit papers on related research to the RoboCup Symposium, which accompanies the annual RoboCup World Championship.

2.10 Desired Abilities

The following is a list of desired technical abilities that the tests in @Home are focusing on:

- Navigation in dynamic environments
- Fast and easy calibration and setup (the ultimate goal is to have a robot up and running out of the box)
- Object recognition
- Object manipulation
- Detection and recognition of humans
- Natural human-robot interaction
- Speech recognition
- Gesture recognition
- Robot applications (@Home is aiming for applications of robots in daily life)
- Ambient intelligence, such as communicating with surrounding devices, retrieving information from the internet, etc.

Chapter 3

General Rules and Regulations

These are the general rules and regulations for the competition in the RoboCup@Home league. Every rule in this section can be considered to implicitly include the term *"unless stated otherwise"*. This means that additional or contrary rules, in particular with respect to the specification of tests, have a higher priority than those mentioned in the general rules and regulations.

3.1 Team Registration and Qualification

3.1.1 Registration and Qualification Process

Each year, there are three phases in the process towards participation in @Home:

1. *Preregistration*
2. *Qualification* announcement
3. Final *Registration* for qualified teams

The preregistration will be announced by a call for participation sent to the *RoboCup@Home mailing list*. Preregistration requires a *team description paper*, a *video* and a *website*. The expected contents of these are described below.

3.1.2 Team Description Paper

The *Team Description Paper* (TDP) is an 8-pages long scientific paper, which must include a description of your main research, including the scientific contribution, goals, scope, and results. The TDP has to be written in English and formatted using the template of the RoboCup International Symposium without any alterations. The paper should contain the following items:

- The focus of research and the contributions in the respective fields
- Innovative technology (if any)
- Re-usability of the system for other research groups
- Applicability of the robot in real-world scenarios
- Photo(s) of the robot(s) used

As an appendix on the 9th page (after the references), please include:

- Team name
- Contact information
- Website URL
- Names of the team members

- Photo(s) of the robot(s), unless included before
- Description of the hardware used
- A brief, compact list of *external devices* (see Section 3.5), if any
- A brief, compact list of any used 3rd party software packages (e.g. ROS' `object_recognition` should be listed, but `OpenCV` doesn't have to be because it is a rather standard library)
- [**Open Platform League only**] A brief description of the hardware used by the robot(s)

During the qualification process, the TDP will be scored according to its scientific value, novelty, and contributions.

3.1.3 Qualification Video

As a proof of running hardware, each team has to provide a *qualification video* that shows at least two of the following abilities (as a minimum requirement):

- Human-robot interaction
- Safe navigation (indoors, with obstacle avoidance)
- Object detection and manipulation
- People detection
- Speech recognition
- Speech synthesis (clear and loud)

Showing some of the following abilities is also recommended:

- Activity recognition
- Complex speech recognition
- Complex action planning
- Gesture recognition
- Failure recovery

The video should not exceed the average time for a test (max. 10 min), should be self-explanatory and designed for a general audience, and should show the robot solving complex tasks relevant for @Home. In particular, to qualify for the competition, the video should show that the robot is able to successfully solve at least one test from the current or previous year's rule book. For robots moving slowly, we suggest to speed-up videos, but please indicate the speed factor being used when doing so (e.g. 2x); the same rule applies for slow motion scenes.

3.1.4 Team Website

The *team website* should be designed for a broader audience and include scientific material (scientific papers, datasets, and documented open source code). The requirements for the website are as follows:

1. **Language**: The team website has to be in English. Other languages may be also available, but English must be default language.
2. **Team**: A comprehensive list of all team members, including brief profiles.
3. **RoboCup**: Link to the league website and previous participations of the team at RoboCup (not necessarily only @Home).

4. **Scientific approach**: Include a research statement, a description of the used approach, and information on scientific achievements.
5. **Publications**: Relevant *publications* from at least the last five years should be included. Downloadable publications are scored higher during the qualification process.
6. **Open source material**: Blueprints, datasets, repositories, or any other kinds of contributions to the league are scored highly during the qualification process.
7. **Multimedia**: Photos and videos of the robot(s) used should be included and easy to find.

3.1.5 Qualification

During the *qualification process*, a selection will be made by the OC. The following points are taken into account and evaluated in the decision process:

- The scientific value, novelty, and contributions of the TDP
- The number of abilities and the complexity of the tasks shown in the qualification video
- The contents of the team website, where relevant publications and open source resources are valued highly

In addition, the following evaluation criteria are considered:

- The performance in previous competitions
- Relevant scientific contributions and publications
- Any additional contributions to the @Home league

Important note for the Standard Platform Leagues: Only unmodified robots may compete in Standard Platform Leagues. Any *unathorized* modification made to the robot that are found in the qualification material will automatically disqualify the team in the qualification process.

3.1.6 Participation Confirmation

In order to have as many participating teams as possible, qualified teams *must* contact the OC to confirm (or cancel) the participation several months in advance. Confirming attendance implies that the team has sufficient resources to register for the competition and commits to attend the event. Qualified teams that do not confirm their participation may be disqualified.

3.2 Audience interaction

Part of making RoboCup@Home appealing is to show the audience what robots should do and what they are actually doing during tasks. In particular, providing information about what a robot is doing to the audience is important for the advancement of the league. To this end, robots in @Home are expected to run the RoboCup@Home VizBox[1], which is a web server to be run on a robot during a task. The page it serves can be displayed on a screen and is visible to the audience via a secondary computer in or around the *Arena*, which is connected to the web server via the wireless network. The *VizBox* can:

[1] https://github.com/LoyVanBeek/vizbox

- display images of what a robot can see, such as camera images, or a visualization of the robot's world model, the robot's map, or anything else that clarifies what the robot is trying to do
- show an outline of the current tasks and the robot's current state in it
- display subtitles of the conversation between a robot and an operator

Additionally, the *VizBox* offers a way to input text commands to the robot so that automatic speech recognition can be bypassed, if necessary.

The documentation of the component is maintained in the *VizBox* repository. All teams should ideally run the same VizBox code, as the audience should be shown a consistent presentation; however, opening a pull request to share any changes is much appreciated so that all teams can benefit from them.

3.3 Scenario

Most competition tests take place in the RoboCup@Home*Arena*, but some tests may take place outside, in a previously unknown public place. In this section, the *Arena* and its contents are described, in particular the furnishing and other information that is common between tests and leagues.

3.3.1 RoboCup@Home Arena

The RoboCup@Home*Arena* is a realistic home setting (an apartment) consisting of interconnected rooms. The minimal configuration consists of:

- a bedroom,
- a dining room,
- a living room, and
- a kitchen

There is usually one *Arena* per league. Depending on the local organization, there may also be multiple *Arena*s that may be different from each other, and a robot must be prepared to perform any task in any *Arena*.

The arena is arranged and decorated to resemble a typical apartment in the hosting country, including all necessities and decorations one can expect to find in a *normal* home. Note that what is considered *normal* may vary greatly based on the culture and location where RoboCup is hosted. Decorations may include, but are not limited to, plants, mirrors, paintings, posters, plates, picture frames, wall clocks, candles with holders, and books.

3.3.2 Walls, Doors, and Floor

The indoor home setting will be surrounded by high and low *walls*, which are built up using standard fair construction material.

1. **Walls:** Walls are fixed and cannot be modified during the competition. The minimum wall height is 60 cm; a maximum height is not specified, but must allow the audience to watch the competition.

2. **Doors:** Inside the *Arena*, rooms are connected by doors (at least one). All doors have handles, not knobs, and can be closed at any time; it is thus expected that robots are either able to open them or find a plan around them. All doors must meet minimum accessibility requirements but they should try to meet the recommended accessibility width of 915 mm.
3. **Floor:** The floor and doorways of the *Arena* are even, so there are no significant steps or stairways; however, minor unevenness, such as carpets, transitions in floor covering between different areas, and minor gaps (especially at doorways) can be expected.
4. **Appearance:** The floor and walls are mostly monochromatic, but may contain textures, such as a carpet on the floor, or a poster or picture on the wall.

3.3.3 Furniture

The *Arena* is furnished with typical objects common for the host country. The minimal configuration consists of:

- a bed,
- a couch,
- a small table,
- a small dinner table with two chairs,
- two trash bins,
- an open cupboard or a small table with a television and remote control,
- a chest of drawers,
- a high cabinet with doors
- a bookcase, and
- a coat rack

The *Arena*'s kitchen has:

- a dishwasher,
- a microwave,
- a sink, and
- a refrigerator (with some cans and plastic bottles inside)

A typical *Arena* setup is shown in Figure 3.1a.

Chest of drawers

The Chest has at least two drawers that are between 90 cm and 120 cm from floor level. The drawers require U-shaped handles.

High Cabinet

The Cabinet can be any shelf-like furniture in which objects can be placed, such that the minimum distance between shelves is 30 cm. The Cabinet needs two side by side doors blocking the access to at least the lower three of the shelves. The doors require U-shaped handles.

Fridge

At least one powered and functioning fridge is required in the *Arena*. The fridge must not be smaller than 120 cm.



(a) Typical Arena **(b)** Typical objects

Figure 3.1: An example of a RoboCup@Home scenario

3.3.4 Changes to the *Arena*

Since robots should be able to function in the real world, the *Arena* is not fixed and might change without further notice.

1. **Major changes:** Any furniture (at a *Predefined Location* or not) that cannot be expected to be fully static in an everyday environment might be moved slightly between tests. In particular, furniture will not change rooms or move drastically inside a room, but a couch or table may be slightly rotated or moved; fixed locations for such furniture items should not be assumed. Walls will stay in place and rooms will not change function. Passages might be blocked.
2. **Minor Changes:** Slightly moved chairs, slightly closed doors, or anything similar cannot be avoided and might happen at any time, even during a test.

3.3.5 Objects

Some tests in the RoboCup@Home league involve recognizing and manipulating objects (see Figure Figure 3.1b). The TC will compile a list of at least 30 objects for this purpose; the list will contain a picture of the each object, as well as its official name and *Object Category* (for instance, an *Apple* belongs to the *Fruits* category). Most objects are likely to be lightweight and easy to grasp with one hand. Every *Object Category* has an assigned *Predefined Location*, where objects of that category can usually be found during tests (for example, an *Fruits* can be found on the *Kitchen Table*); assignments are announced during the *Setup Days* (see Section 4).

Objects are provided at the competition for training. Teams may keep at most five training objects at a time and for at most one hour. Modifying the training objects is not allowed.

Two types of objects are used in the tasks:

1. ***Known Objects*:** Objects previously known to the robot, divided into:
   1.1. ***Consistent Objects*:** Objects whose image appears in the list of objects.
   1.2. ***Similar Objects*:** Objects whose image is not present in the list of objects, but look similar enough to one of them that a person would consider them the same kind of object. For example, an apple whose color is different from the apple in the list of objects, or a piece of cloth with a different pattern.

1.3. ***Standard Objects*:** Objects chosen from the *YCB Dataset*.[2] They are published 6 months in advance on the RoboCup@Home website[3], so that they can be aquired and trained beforehand.

2. ***Unknown Objects*:** Any other object that is not in the object list but can be grasped or handled (e.g., *Arena* decorations).

The minimal configuration of *Known Objects* consists of:

- ***Tableware*:** Dish, bowl, cup (or mug), and napkin (see Figure 3.2a).
- ***Cutlery*:** Fork, knife, and spoon.
- ***Trash Bags*:** Big plastic trash bags, preferably with a handle.
- ***Bags*:** Lightweight and with stiff, vertical handles (see Figure 3.2b).
- ***Dry Food Container*:** Dry food containers (see Figure 3.2d).
- ***Disks or books*:** A set of discs (LP, CD, DVD, or BluRay) or books.
- ***Coat rack*:** A rack or pole to hang coats and other clothes.
- ***Trays*:** A transport object such as a tray or basket, intended for bimanual manipulation (see Figure 3.2c).
- ***Pourable*:** An object whose content can be poured (such as a cereal box).
- ***Heavy object*:** Weight between 1.0kg and 1.5kg.
- ***Tiny object*:** A lightweight object that is not bigger than 5 cm (such as paper, a teabag, or a pen).
- ***Fragile object*:** An easy-to-break object, (such as a chocolate egg).
- ***Deformable object*:** A flexible object that may take different shapes (such as cloth).
- ***Garbage bag*:** A garbage bag that can be tied.



(a) Cereal bowls **(b)** Bright-colored paper bags **(c)** Serving tray **(d)** Dry

Figure 3.2: Example of object containers

During the competition, objects can be requested based on their *Object Category*, physical attributes, or a combination of both. Relevant attributes to be used are:

[2] http://www.ycbbenchmarks.com/object-set/
[3] https://athome.robocup.org/standard-objects

- Color (such as red, blue, black with white dots, etc.).
- Relative estimated size (smallest, largest, big one, etc.).
- Relative estimated weight (lightest, heaviest).
- Relative position (left of, rightmost, etc.).
- Object description (is fragile, is a container, can be poured, requires two hands, etc.).

Remark: Measurements are estimations and based on common sense. It is OK for robots to consider similar objects to be about the same size or weight.

3.3.6 Predefined Rooms and Locations

Some tests in the RoboCup@Home league involve a *Predefined Location* where people or objects can be found. There will also be at least two *doors*, named an *Entrance* and an *Exit*, which lead in and out of the *Arena*, respectively. Room names, predefined locations, and location classes are announced during the *Setup Days* (see Section 4).

3.3.7 Predefined Person Names

Some tests in the RoboCup@Home league involve memorizing a person's name. All people in the *Arena* have an assigned *Predefined Name* chosen by the TC. The list of names contains 25 % male, 25 % female, and 50 % gender-neutral names taken from the list of most commonly used names in the United States. Predefined names are announced during the *Setup Days* (see Section 4).

3.3.8 Wireless network

For wireless communication, an *Arena Network* is provided. The actual infrastructure depends on the local organization. Reliability and performance of the network is not guaranteed; robots are expected to be able to run without a wireless network.

The following rules apply:

- Only the *Arena Network* can be used during tests.
- Only the active team in a task is allowed to use the *Arena Network*.
- The *Arena Network* provides one Virtual Local Area Network (VLANs) per team.
- Each VLAN is most likely to have its own SSID/password.
- VLAN traffic is separated from any other team and is routed to the team's network cable in the team area.
- Each VLAN is also connected to the Internet.

Teams broadcasting unauthorized (aka rogue) wireless networks will be disqualified from the competition and their devices may be temporarily confiscated by the OC, this includes smartphones and concealed SSIDs. It is thus advised to verify your devices for any breaches of this nature.

3.4 Robots

3.4.1 Number of Robots

1. **Registration:** The maximum *number of robots* per team is *two* (2).

2. **Regular Tests:** Only one robot is allowed per test. For different test runs, different robots can be used.
3. ***Final*:** In the *Final*, both robots can be used simultaneously.

3.4.2 Appearance and Safety

Robots should have a product-like appearance and be safe to operate. The following rules apply to all robots:

1. **Cover:** The robot's internal hardware (electronics and cables) should be covered so that safety is ensured. The use of (visible) duct tape is strictly prohibited.
2. **Loose cables:** Loose cables hanging out of the robot are not permitted.
3. **Safety:** The robot must not have sharp edges or elements that might harm people.
4. **Annoyance:** The robot must not be continuously making loud noises or use blinding lights.
5. **Marks:** The robot may not exhibit any kind of artificial marks or patterns.
6. **Driving:** To be safe, the robots should be careful when driving. Obstacle avoidance is mandatory.

The compliance with these rules will be verified during *Robot Inspection* (see 4.4).

3.4.3 Standard Platform Leagues

Standardized platforms allow teams to compete under equal conditions by eliminating all hardware-dependent variables from the tasks; therefore, *unauthorized* modifications and alterations to the robots are strictly forbidden. This includes, but is not limited to, attaching, connecting, plugging, gluing, and taping components into and onto the robot, as well as, modifying or altering the robot structure. Not complying with this rule leads to an immediate disqualification and penalization of the team (see Section 3.9.2). Robots are, however, allowed to "wear" clothes, have stickers (such as a sticker exhibiting the logo of a sponsor), or be painted (provided that the robot provider has approved that).

All modifications to the robots will be examined during the *Robot Inspection* (see 4.4). Note that the EC and TC members may request re-inspection of an SPL robot at any time during the competition.

Authorized DSPL Modifications

In the *Domestic Standard Platform League* (DSPL), teams may use an external laptop, which is safely located in the official *Mounting Bracket* provided by Toyota and is connected to the *Toyota HSR* via an Ethernet cable. Any laptop fitting inside the *Mounting Bracket* is allowed to be used, regardless of its technical specification. Furthermore, teams are allowed to attach the following devices to either the *Toyota HSR* or the laptop in the *Mounting Bracket*:

- **Audio**: A USB audio output device, such as a USB speaker or a sound card dongle.
- **Microphone**: A USB or AUX external microphone. Wireless microphones are not allowed.
- **Wi-Fi adapter**: A USB-powered IEEE 802.11ac (or newer) compliant device.

- **Ethernet Switch**: A USB-powered IEEE 802.3ab (or newer) compliant device.

In all cases, a maximum of three such devices can be attached, such that they may not increase the robot's dimensions. For this purpose, using short cables and attaching the devices to the laptop in the *Mounting Bracket* is advised.

3.4.4 Robot Specifications for the Open Platform League

Robots competing in the RoboCup@Home Open Platform League must comply with security specifications in order to avoid causing any harm while operating.

Size and Weight

1. **Dimensions:** The dimensions of a robot should not exceed the limits of an average door (200 cm by 70 cm in most countries). The TC may allow the qualification and registration of larger robots, but, due to local restrictions, it cannot be guaranteed that the robots can actually enter the *Arena*. In doubt, please contact the *Local Organizing Committee* (LOC).
2. **Weight:** There are no specific weight restrictions; however, the weight of the robot and the pressure it exerts on the floor should not exceed local regulations for the construction of offices and/or buildings which are used for living in the country where the competitions is being held.
3. **Transportation:** Team members are responsible for quickly moving the robot out of the *Arena*. If the robot cannot move by itself (for any reason), the team members must be able to transport the robot away quickly and easily.

Appearance

OPL robots should have an appearance that resembles a safe and finished product rather than an early stage prototype. This, in particular, means that the robot's internal hardware (electronics and cables) should be completely covered so that safety is ensured. Please note that covering the robot's internal hardware with a t-shirt is not forbidden, but is not advised.

Emergency Stop Button

1. **Accessibility and visibility:** Every robot has to provide an easily accessible and visible *Emergency Stop* button.
2. **Color:** The *Emergency Stop* must be coloured red and be the only red button on the robot. The TC may ask the team to tape over or remove any other red buttons present on the robot.
3. **Robot behavior:** When the *Emergency Stop* button is pressed, the robot and all its parts must stop moving immediately.

Start Button

1. **Requirements:** As explained in Section 3.7.8, teams that aren't able to carry out the default start signal (opening the door) have to provide a *Start Button* that can be used to

start tests. Teams need to announce this to the TC before every test that involves a start signal, including the *Robot Inspection*.

2. **Definition:** The *Start Button* can be any "one-button procedure" that can be easily executed by a referee (such as releasing the *Emergency Stop*, a green button, or a software button in a graphical user interface).

Note: All robot requirements will be tested during the *Robot Inspection* (see 4.4).

3.5 External devices

Everything that a team uses during a test and is not part of the robot is considered an *External Device*. All *External Devices* must be authorized by the *technical committee* (TC) during the *Robot Inspection* test (see Section 4.4). The TC specifies whether an *External Device* can be used freely or under referee supervision, and determines its impact on scoring.

Note that the use of wireless devices, such as hand microphones and headsets, is not allowed, with the exception of *External Computing* as specified below. Please also note that the competition organizers do not guarantee or take any responsibility regarding the availability or reliability of the network or the internet connection in the *Arena*. Teams can thus use *External Computing* resources at their own risk.

3.5.1 On-site External Computing

Computing resources that are not physically attached to the robot are considered *External Computing* resources. The use of up to five *External Computing* resources is allowed, but only in the *Arena Network* (see Section 3.3.8) and with a prior approval of the TC. Teams must inform the TC about the use of any *External Computing* at least one month before the competition. Note, however, that robots must be able to operate safely even if *External Computing* is unavailable.

External Computing devices must be placed in the *External Computing Resource Area* (ECRA), which is announced by the TC during the *Setup Days*. A switch connected to the *Arena* wireless network will be available to teams in the ECRA. During a *Test Block*, at most two laptops and two people from different teams are allowed in the ECRA simultaneously, one member each of the teams up next. No peripherals, such as screens, mice, keyboards, and so forth, are allowed to be used.

During a *Test Slot*, everyone must stay at least 1 m away from the ECRA. Interacting with anything in the ECRA after the referee has given the start signal for a test will result in the test being stopped with a score of zero.

If a laptop is used as *External Computing*, a team can only place it in the ECRA if their *Test Slot* is up next and must remove the device immediately after the test.

3.5.2 On-line external computing

Teams are allowed to use *External Computing* through the internet connection of the *Arena Network*; this includes cloud services or online APIs. These must be announced to and approved by the TC one month prior to the competition.

3.6 Organization of the Competition

3.6.1 Stage System

The competition features a *stage system*, namely it is organized in two stages, each consisting of a number of specific tasks, and ends with a *Final*.

1. **Robot Inspection:** For security, robots are inspected during the *Setup Days*. A robot must pass the *Robot Inspection* test (see Section 4.4) so that it is allowed to compete.
2. ***Stage I*:** The first days of the competition are called *Stage I*. All qualified teams that have passed the *Robot Inspection* can participate in *Stage I*.
3. ***Stage II*:** The best *50% of teams* after *Stage I* advance to *Stage II*. If the total number of teams is less than 12, up to 6 teams may advance to *Stage II*. In this stage, tasks require more complex abilities or combinations of abilities.
4. ***Final*:** The best *two teams* of each league, namely the ones with the highest score after *Stage II*, advance to the *Final*. The final round features only a single task integrating all tested abilities.

In case of having no considerable score deviation between a team advancing to the next stage and a team dropping out, the TC may announce additional teams advancing to the next stage.

3.6.2 Schedule

1. **Test Blocks:** For each test one Test Block is scheduled during which each (qualified) team gets at least two Test Slots assigned. With the Restaurant test being the exception where only one Test Slot is assigned.
2. **Test Slots:** During a Test Slot a team has the amount of time specified for the test to attempt to solve it.
3. **Participation is default:** Teams have to inform the OC in advance if they are skipping a Test Block. Without such indication, they may receive a penalty when not attending (see Section 3.9.1).

	Day 1	Day 2	Day 3	Day 4
Block 1 (9:00–11:00)	Help Me Carry	Serve Breakfast	Restaurant	
	—Break—			**Finals**
Block 2 (13:00–15:00)	Receptionist	GPSR	Stickler for the Rules	
	Stage 2			
Block 3 (17:00–19:00)	Storing Groceries	Clean the Table	EGPSR	

Table 3.1: Example schedule. Each team has at least two Test Slots assigned in every Test Block.

Note that the actual allocation of blocks will be announced by the OC during the *Setup Days* (see Table 3.1).

3.6.3 Scoring System

Each task has a main objective and a set of bonus scores. To score in a test, a team must successfully accomplish the main objective of the task; bonuses are not awarded otherwise.

The scoring system has the following constrains:

1. ***Final*:** The final score is normalized.
2. **Minimum score:** The minimum total score per test in *Stage I* and *Stage II* is *0 points*. In principle, teams cannot receive negative points, except if they receive penalties. In particular, both penalties for not attending (see Section 3.9.1) and extraordinary penalties (see Section 3.9.2) can result in a total negative score.

3.7 Procedure during Tests

3.7.1 Safety First!

1. **Emergency Stop:** At any time when operating the robot inside and outside the scenario the owners have to stop the robot immediately if there is a possibility of dangerous behavior towards people and/or objects.
2. **Stopping on request:** If a referee, member of the Technical or Organizational committee, an Executive or Trustee of the federation stops the robot (by pressing the emergency button) there will be no discussion. Similarly if they tell the team to stop the robot, the robot must be stopped *immediately*.
3. **Penalties:** If the team does not comply, the team and its members will be excluded from the ongoing competition immediately by a decision of the RoboCup@Home *Technical Committee* (TC). Furthermore, the team and its members may be banned from future competitions for a period not less than a year by a decision of the RoboCup Federation Trustee Board.

3.7.2 Maximum number of team members

1. **Regular Tests:** During a regular test, the maximum number of team members allowed inside the *Arena* is *one* (1). Exceptions are tests that explicitly require volunteer assistance.
2. **Setup:** During the setup of a test, the number of team members inside the *Arena* is not limited.

3.7.3 Fair play

Fair Play and cooperative behavior is expected from all teams during the entire competition, in particular:

- while evaluating other teams,
- while refereeing, and
- when having to interact with other teams' robots.

This also includes:

- not trying to cheat (e.g., pretending autonomous behavior where there is none),
- not trying to exploit the rules (e.g., not trying to solve the task but trying to score), and
- not trying to make other robots fail on purpose.
- not modifying robots in standard platforms.

Disregard of this rule can lead to penalties in the form of negative scores, disqualification for a test, or even for the entire competition.

3.7.4 Expected Robot's Behavior

Unless stated otherwise, it is expected that the robot always behave and react in the same way a polite and friendly human being would do. This applies also to how robots try solve the assigned task As rule of thumb, one may ask any non-scientist how she would solve the task.

Please consider that average users will not know the specific procedure to operate a robot. Hence, interaction should be as with any other human being.

3.7.5 Robot Autonomy and Remote Control

1. **No touching:** During a test, the participants are not allowed to make contact with the robot(s), unless it is in a "natural" way and required by the task.
2. **Natural interaction:** The only allowed means to interact with the robot(s) are gestures and speech.
3. **Natural commands:** Anything that resembles direct control is forbidden.
4. **Remote Control:** Remotely controlling the robot(s) is strictly prohibited. This also includes pressing buttons, or influencing sensors on purpose.
5. **Penalties:** Disregard of these rules will lead to disqualification for a test or for the entire competition.

3.7.6 Collisions

1. ***Touching*:** Gently *touching* objects is tolerated but unadvised. However, robots are not allowed to crash with something. The "safety first" rule (Section 3.7.1) overrides any other rule.
2. ***Major collisions*:** If a robot crushes into something during a test, the robot is immediately stopped. Additional penalties may apply.
3. ***Functional touching*:** Robots are allowed to apply pressure on objects, push away furniture and, in general, interact with the environment using structural parts other than their manipulators. This is known as *functional touching*. However, the robot must clearly announce the collision-like interaction and kindly request not being stopped.
   Remark: Referees can (and will) immediately stop a robot in case or suspicion of *dangerous* behavior.

3.7.7 Removal of robots

Robots not obeying the rules are stopped and removed from the *Arena*.

1. It is the decision of the referees and the TC member monitoring the test if and when to remove a robot.

2. When told to do so by the referees or the TC member monitoring the test, the team must immediately stop the robot, and remove it from the *Arena* without disturbing the ongoing test.
3. More than 1 team member is allowed to enter the *Arena* after the robot has been stopped to quickly remove the robot from the *Arena*

3.7.8 Start Signal

The default *start signal* (unless stated otherwise) is *door opening*. Other start signals are allowed but must be authorized by the *Technical Committee* (TC) during the Robot Inspection (see Section 4.4).

1. **Door opening:** The robot is waiting behind the door, outside the *Arena* and accompanied by a team member. The test starts when a referee (not a team member) opens the door.
2. **Start button:** If the robot is not able to automatically start after the door is open, the team may start the robot using a start button.
   2.1. It must be a physical button on the robot (e.g., a dedicated one or releasing the eStop).
   2.2. It is allowed to use the robot's contact/pressure sensors (e.g., pushing the head or an arm joint).
   2.3. Using a start button needs to be announced to the referees before the test starts.
   2.4. There may be penalties for using a start button in some tests.
3. **Ad-hoc start signal:** Other means of triggering the robot to action are allowed but must be approved by the *Technical Committee* (TC) during the Robot Inspection (see Section 4.4). These include:
   - QR Codes
   - Verbal instructions
   - Custom HRI interfaces (apps, software, etc.)

   Remark: There may be penalties for using Ad-hoc start signals in some tests. The use of mouses, keyboards, and devices attached to ECRA computers is strictly forbidden.
4. **No penalties for physical constraints (HSR only):** In cases where a physical issue, such as a door bump, impedes the league's ability to ensure a standard start signal with full safety for the HSR robot, penalties will not be imposed. The following guidelines apply:
   - The team must document the physical issue and notify the committee prior to the Robot Inspection.
   - The issue must be reviewed and approved by the league's rules committee.
   - An alternative start method will be determined on-site in consultation with the committee.

   This rule ensures fairness and flexibility while addressing unavoidable physical challenges during league operations.

3.7.9 Restart Rule

During any test, teams are allowed to perform a single restart per attempt under the following conditions:

- A restart may only be requested until the robot completes a major scoring action.
- The test time will continue running until the team has exited the *Arena* with the robot.
- A restart incurs a penalty of 30 s, subtracted from the total time.
- After exiting, the team may requeue at the back of the line and begin their attempt again within their remaining time.
- Only one restart is permitted per test attempt.
- Only points scored after a restart are counted; points scored before the restart are not considered if a restart was requested.

3.7.10 Entering and leaving the *Arena*

1. **Start position:** Unless stated otherwise, the robot starts outside of the *Arena*.
2. **Entering:** The robot must autonomously enter the *Arena*.

3.7.11 Gestures

Hand gestures may be used to control the robot in the following way:

1. **Definition:** The teams define the hand gestures by themselves.
2. **Approval:** Gestures need to be approved by the referees and TC member monitoring the test. Gestures should not involve more than the movement of both arms. This includes, e.g., expressions of sign language or pointing gestures.
3. **Instructing operators:** It is the responsibility of the team to instruct operators.
   3.1. The team may only instruct the operator when told to so by a referee.
   3.2. The team may only instruct the operator in the presence of a referee.
   3.3. The team may only instruct the robot for as long as allowed by the referee.
   3.4. When the robot has to instruct the operator, it is the robot that instructs the operator and *not* the team. The team is not allowed to additionally guide the operator, e.g., tell the operator to come closer, speak louder, or to repeat a command.
   3.5. The robot is allows to instruct the operator at any time.
4. **Receiving gestures:** Unless stated otherwise, it is not allowed to use a speech command to set the robot into a special mode for receiving gestures.

3.7.12 Referees

All tests are monitored by a referee, who is a member of the *Technical Committee* (TC). The referee may appoint an assistant to aid in keeping time and filling in scoresheets. The following rules apply:

1. **Selection:** Referees are chosen by EC/TC/OC.
2. **Referee instructions:** Right before each test, referee chooses one or more assistant to aid during the test. The assistants will be instructed by the referee.

3.7.13 Operators

Unless stated otherwise, robots are operated by the referee or by a person selected by the referee. If the robot fails to understand the default operator, the team may request the use of a custom operator. Penalty may apply when using a custom operator.

3.7.14 Time limits

1. **Stage I:** Unless stated otherwise, the time limit for each test in *Stage I* is *5 minutes*.
2. **Stage II:** Unless stated otherwise, the time limit for each test in *Stage II* is *10 minutes*.
3. **Inactivity:** Robots are not allowed to stand still or get stuck into endless loops. A robot not progressing in the task execution (and obviously not trying to), is consider as inactive. Robots must be removed after 30 seconds of inactivity.
4. **Requesting time:** A robot (not the team) can request referees to make exception from the 30-seconds inactivity time limit. In its request, the robot must clearly state for how long it will be performing a time-consuming process (e.g., 60 seconds). This time cannot exceed 3 minutes and cannot be used more than once per test.
5. **Setup time:** Unless stated otherwise, there is no setup time. Robots need to be ready to enter the *Arena* no later than one minute after the door has been closed to the former team.
6. **Time-up:** When the time is up, the team must immediately remove their robot(s) from the *Arena*. No more additional score will be giving.
7. **Show must go on:** On special cases, the referee may let the robot continue the test for demonstration purposes, but no additional points will be scored.

3.8 Deus Ex Machina: Bypassing Features With Human Help

Because the Show Must Go On

Robots can't score unless they accomplish the main goal of a task. However, in many real-life situations, a minor malfunction may prevent the robot from accomplishing a task. To prevent this situation, while fostering awareness and human-robot interaction, robots are allowed to request human assistance during a test.

3.8.1 Procedure

The procedure to request human assistance while solving a task is as follows:

1. **Request help:** The robot must indicate loud and clear that it requires human assistance. It must be clearly stated:
   - The nature of the assistance
   - The particular goal or desired result
   - How the action must be carried out (when necessary)
   - Details about how to interact with the robot (when necessary)
   - Detailed information to identify objects for picking and placing (e.g. object name, color or location). The provided information needs to show that objects were perceived by the robot.

2. **Supervise:** The robot must be aware of the human's actions, being able to tell when the requested action has been completed, as well as guiding the human assistant (if necessary) during the process.
3. **Acknowledge:** The robot must politely thank the human for the assistance provided.

Example

In the following example, a robot has to clean the table but is unable to grasp the spoon.

R: `I am sorry, but the spoon is too small for me to take.`
`Could you please help me with it?`
`Please say "robot yes" or "robot no" to confirm.`
H: *Robot, yes!*
R: `Thank you! Please follow my instructions.`
`Please take the purple spoon from the table. It is on my left.`
`(The robot also shows the result of the perception, e.g. by pointing at the spoon or showing a picture with a bounding box on the screen)`
H: (Referee takes purple spoon)
R: `I saw you took the spoon.`
`Would you be so kind as to follow me to the kitchen?`
`Please keep the spoon visible in front of you so I can track you. Thank you!`
R: `You can stop following me now.`
`As you can see, the dishwasher is already open.`
`Please place the spoon in the gray basket on the lower tray.`
R: `Lovely! Thanks for your help.`
`I'll let you know if I need further assistance.`

3.8.2 Scoring

There is no limit in the amount of times a robot can request human assistance, but score reduction applies every time it is requested.

1. **Partial execution:** A reduction of 10% of the maximum attainable score is applied when the robot request a partial solution (e.g. pointing to the person the robot is looking for or placing an object within grasping distance). The referee decides whether the requested action is simple enough to corresponds to a partial execution or not.
2. **Full awareness:** A reduction of 20% of the maximum attainable score is applied when the robot is able to track and supervise activity, detecting possible, and when the requested action has been completed.
3. **No awareness:** A reduction of 30% of the maximum attainable score is applied when the robot has to be told when the requested action has been completed.
4. **Bonuses:** No bonus points can be scored when the robot requests help to solve part of a task that normally would grant a bonus.
5. **Score reduction overlap:** The score reduction for multiple requests of the same kind do not stack, but overlap. The total reduction applied correspond to the worse execution (higher reduction of all akin help requests). This means, a robot won't be reduced again for requesting help to transport a second object, but a second reduction will apply when the robot asks for a door to be opened.

6. **Allowed types of assistance:** The types of assistance allowed in a given task are specified in the respective task description. It should be noted that only the assistance types explicitly mentioned in a task description are actually allowed in a task; other types of assistance are not allowed and will nullify the obtained points for the part of the task in which they are applied. For instance, if a task focused on manipulation does not explicitly mention a Deus Ex Machina penalty for instructing a person to perform a manipulation activity, it should not be assumed that this is a loophole that can be exploited.

3.8.3 Bypassing Automatic Speech Recognition

Giving commands to the robot is essential in many tests. When the robot is not able to receive spoken commands, teams are allowed to provide means to bypass ASR via an Alternative method for HRI (see Section 3.8.3). Nonetheless, Automatic Speech Recognition is preferred.

The following rules apply in addition to the ones specified in section Section 3.8.2

1. **ASR with Default Operator:** No score reduction. The command is given by the human operator who must speak (not shout) loud and clear. The *default operator* may repeat the command up to three times.
2. **ASR with Custom Operator:** A reduction of 10% of the maximum attainable score is applied when a *custom operator* is requested. The Team Leader chooses a person who gives the command *exactly as instructed by the referee.*
3. **Gestures:** A reduction of 20% of the maximum attainable score is applied when a gesture (or set of gestures) is used to instruct the robot.
4. **QR Codes:** A reduction of 30% of the maximum attainable score is applied when a QR code is used to instruct the robot.
5. **Alternative Input Method:** A reduction of up to 30% of the maximum attainable score is applied when a *alternative HRI interface*, is used to instruct the robot. Alternative HRI interfaces (see Section 3.8.3) must be previously approved by the TC during the Robot Inspection (see Section 4.4).

Alternative interfaces for HRI

Alternative methods and interfaces for HRI offer a way for a robot to start or complete a task. Any reasonable method may be used, with the following criteria:

- **Intuitive to use and self-explanatory:** a manual should not be needed. Teams are not allowed to explain how to interface with the robot.
- **Effortless use:** Must be as easy to use as uttering a command.
- **Is smart and preemptive:** The interface adapts to the user input, displaying only the options that make sense or that the robot can actually perform.
- Exploits the best of the device being used (eg. touch screen, display area, speakers, etc.)

Preferably, the alternative HRI must be also adapted to the user. Consider localization (with English as the default), but also potential users of service robots at their home. For example: elderly people and people with physical disabilities.

3.9 Special penalties and bonuses

3.9.1 Penalty for not attending

1. **Automatic schedule:** All teams are automatically scheduled for all tests.
2. **Announcement:** If a team cannot participate in a test (for any reason), the team leader has to announce this to the OC at least *60 minutes* before the test slot begins.
3. **Penalties:** A team that is not present at the start position when their scheduled test starts, the team is not allowed to participate in the test anymore. If the team has not announced that it is not going to participate, it gets a penalty of *250 points*.

3.9.2 Extraordinary penalties

1. **Penalty for cheating:** If a team member is found cheating or breaking Fair Play, the team will be automatically disqualified of the running test, and a penalty of *500 points* is handed out. The *Technical Committee* (TC) may also disqualify the team for the entire competition.
2. **Penalty for faking robots:** If a team starts a test, but it does not solve any of the partial tasks (and is obviously not trying to do so), a penalty of *250 points* is handed out. The decision is made by the referees and the monitoring TC member.
3. **Extra penalty for collision:** In case of major, (grossly) negligent collisions the *Technical Committee* (TC) may disqualify the team for a test (the team receives *0 points*), or for the entire competition.
4. **Not showing up as referee or jury member:** If a team does not provide a referee or jury member (being at the *Arena* on time), the team receives a penalty of *250 points*, and will be remembered for qualification decisions in future competitions.
   Jury members missing a performance to evaluate are excluded from the jury, and the team is disqualified from the test (receives *0 points*).
5. **Modifying or altering standard platform robots:** If any unauthorized modification is found on a Standard Platform League robot, the responsible team will be immediately disqualified for the entire competition while also receiving a penalty of *500 points* in the overall score. This behavior will be remembered for qualification decisions in future competitions.

3.9.3 Bonus for outstanding performance

1. For every regular test in *Stage I* and *Stage II*, the @Home *Technical Committee* (TC) can decide to give an extra bonus for *outstanding performance* of up to 10% of the maximum test score.
2. This is to reward teams that do more than what is needed to solely score points in a test but show innovative and general approaches to enhance the scope of @Home.
3. If a team thinks that it deserves this bonus, it should announce (and briefly explain) this to the *Technical Committee* (TC) beforehand.
4. It is the decision of the *Technical Committee* (TC) if (and to which degree) the bonus score is granted.

3.10 Open Challenge

On the first two competition days after the end of the regular test blocks, teams will have an opportunity to present an *Open Challenge* in which they demonstrate their novel research and approaches.

3.10.1 Procedure

1. **Participation:** Teams have to announce whether they want to perform an *Open Challenge* to the OC during the *Setup Days*.
2. **Time:** Each team gets a 10 minute time slot for the *Open Challenge*; of these, 8 minutes are for presenting and 2 minutes are for questions by the audience.
3. **Arena Changes:** The team can rearrange the arena when their time slot starts, but all changes need to be reverted as soon as their time slot ends.
4. **Focus:** While the demonstrations are intended to share research insights, we still want to see robots performing; in particular, the *Open Challenge* should not be turned into a pure academic lecture.
5. **Leagues:** Ideally, the open challenges of all @Home leagues will be scheduled consecutively so that everyone has an opportunity to see all demonstrations; however, in case more than 12 participants across the leagues register for the *Open Challenge*, each league will hold their *Open Challenge* concurrently.
6. **Award:** The *Open Challenge* does not contribute any points towards the official competition score, but participating teams are eligible to receive the *Best Open Challenge Award* (see 1.7.4).

Chapter 4

Setup and Preparation

Prior to the RoboCup@Home competition, all arriving teams will have an opportunity to set up their robots and prepare for the competition in a *Setup & Preparation* phase. This phase is scheduled to start on the first day of the competition, namely when the venue opens and the teams arrive. During the *Setup Days*, teams can assemble and test their robots. On the last setup day, a *Welcome Reception* will be held. To foster the knowledge exchange between teams a conference-like *Poster Session* takes place during the reception. Additionally, all teams have to get their robots inspected by members of the TC to be allowed to participate in the competition.

Regular tests are not conducted during the setup & preparation phase. The competition starts with *Stage I* (see Section 5).

Table 4.1: Stage System and Schedule per League (distribution of tests and stages over days may vary)

Setup & Preparation	*Stage I*	*Stage II*	*Finals*
	advance → All teams that passed Inspection	*advance* → Best 6 (< 12) or best 50% (≥ 12)	*advance* → Best 3 teams

4.1 General Setup

Depending on the schedule, the *Setup & Preparation* phase lasts for one or two days.

1. **Start:** The *Setup & Preparation* starts when the venue opens for the first time.
2. **Intention:** During the *Setup & Preparation*, teams arrive, bring or receive their robots, and assemble and test them.
3. **Tables:** The local organization will set up and randomly assign team tables.
4. **Groups:** Depending on the number of teams, the *organizing committee* (OC) may form multiple groups of teams (usually two) for the first (and second stage). The OC will assign teams to groups and announce the assignment to the teams.
5. ***Arena*:** The *Arena* is available to all teams during the *Setup & Preparation*. The OC may schedule special test or mapping slots in which *Arena* access is limited to one or more teams exclusively (all teams get slots). Note, however, that the *Arena* may not yet be complete and that the last work is conducted in the *Arena* during the *Setup Days*.

6. **Objects:** The delegation of EC, TC, OC and local organizers will buy the objects (see Section 3.3.5). Note, however, that the objects may not be available at all times and not from the beginning of the *Setup & Preparation*.

4.2 Welcome Reception

Since Eindhoven 2013, RoboCup@Home holds an own *Welcome Reception* in addition to the official opening ceremony. During the *Welcome Reception*, a *Poster Session* is held in which teams present their research focus and latest results (see Section 4.3).

1. **Time:** The *Welcome Reception* is held in the evening of the last setup day.
2. **Place:** The *Welcome Reception* takes place in the @Home *Arena* and/or in the @Home team area.
3. **Snacks & drinks:** During the *Welcome Reception*, snacks and beverages (beers, sodas, etc.) are served.
4. **Organization:** It is the responsibility of the OC and the local organizers to organize the *Welcome Reception* and *Poster Session*, including:
   4.1. organizing poster stands (one per team) or alternatives for presenting the posters,
   4.2. organizing snacks and drinks, and
   4.3. inviting officials, sponsors, the local organization, and the trustees of the RoboCup Federation to the event.
5. **Poster presentation:** During the *Welcome Reception*, the teams give a poster presentation on their research focus, recent results, and their scientific contribution. Both the poster and the teaser talk are evaluated by a jury (see Section 4.3).

4.3 Poster Teaser Session

Before the *Welcome Reception* and *Poster Session*, a *Poster Teaser Session* is held. In this teaser session, each team can give a short presentation of their research and the poster being presented at the poster session.

4.3.1 Poster teaser session

1. **Presentation:** Each team has a maximum of three minutes to give a short presentation of their poster.
2. **Time:** The *Poster Teaser Session* is to be held before the *Welcome Reception* and *Poster Session* (see Section 4.2).
3. **Place:** The *Poster Session* may be held in or around the *Arena*, but should not interfere with the *Robot Inspection* (see Section 4.4).
4. **Evaluation:** The teaser and poster presentations are evaluated by a jury consisting of members of the other teams. Each team has to provide one person (preferably the team-leader) to follow and evaluate the entire *Poster Teaser Session* and the *Poster Session*.
5. **Criteria:** For each of the following evaluation criteria, a maximum of 10 points is given per jury member:

5.1. Novelty and scientific contribution
5.2. Relevance for RoboCup@Home
5.3. Presentation (quality of poster, teaser talk, and discussion during the *Poster Session*)

6. **Score:** The points given by each jury member are scaled to obtain a maximum of 50 points. The total score for each team is the mean of the jury member scores. To neglect outliers, the N best and worst scores are left out:

$$score = \frac{\sum \text{team-leader-score}}{\text{number-of-teams} - (2N+1)}, N = \begin{cases} 1, & \text{number-of-teams} \geq 10 \\ 2, & \text{number-of-teams} < 10 \end{cases}$$

7. **Sheet collection:** The evaluation sheets are collected by the OC at a later time (announced beforehand by the OC), allowing teams to continue knowledge exchange during the first days of the competition (*Stage I*).
8. **OC Instructions:**
   - Prepare and distribute evaluation sheets before the *Poster Teaser Session*.
   - Collect the evaluation sheets.
   - Organize and manage the poster teaser presentations and the *Poster Session*.

4.4 Robot Inspection

Safety is the most important issue when interacting with humans and operating in the same physical workspace. Because of this, all participating robots are inspected before participating in RoboCup@Home. Every team needs to get their robot(s) inspected and approved so that they can participate in the competition.

1. **Procedure:** The *Robot Inspection* is conducted like a regular test, namely it starts with opening of the arena door (see Section 3.7.8). One team after another (and one robot after another) has to enter the *Arena* through a designated entrance door, move to the *Inspection Point*, and leave the arena through the designated exit door. In between entering and leaving, the robot is inspected by the *technical committee* (TC).
2. **Checked aspects:** During the *Robot Inspection*, each robot is checked for compliance with the competition rules (see Section 3.4), in particular:
   - emergency button(s)
   - collision avoidance (a TC member steps in front of the robot)
   - voice of the robot (it must be loud and clear)
   - custom containers (bowl, tray, etc.)
   - external devices (including wireless network), if any
   - Alternative human-robot interfaces (see Section 3.8.3).
   - ***Standard Platform robots***
     - no modifications have been made
     - specification of the *Official Standard Laptop* (OSL) (if required)
   - ***Open Platform robots***
     - robot speed and dimension
     - start button (if the team requires it)

  - robot speaker system (plug for RF transmission)
  - other safety issues (duct tape, hanging cables, sharp edges etc.)

3. **Re-inspection:** If the robot is not approved in the inspection, it is the responsibility of the team to get the approval at a later point. Robots are not allowed to participate in any test before passing the *Robot Inspection*.
4. **Time limit:** The robot inspection is interrupted after three minutes (per robot). When told so by the TC — in case of time interrupt or failure — the team has to move the robot out of the *Arena* through the designated exit door.
5. **Appearance Evaluation:** In addition to the inspection, the TC evaluates the appearance of the robots. Robots are expected to look nice (no duct tape, no cables hanging loose etc.). In case of objection, the TC may penalize the team with a penalty of maximum 50 points.
6. **Accompanying team member:** Each robot is accompanied by only one team member (the team leader is advised).
7. **OC instructions (at least two hours before the *Robot Inspection*):**
   - Announce the entry and exit doors.
   - Announce the location of the *Inspection Point* in the *Arena*.
   - Specify and announce where and when the poster teaser and the poster presentation session take place.

Chapter 5

Tests in Stage I

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5.1 Help Me Carry

The robot's owner needs help carrying something from the car to the starting location.
Main Goal: The robot helps the operator to carry a bag from a car parked outside.

Optional Goals:
1. Re-reaching the starting location
2. Following the queue on the way back

Focus

Person following, *navigation in unmapped environments*, *social navigation*.

Setup

- **Locations**:
  - The test is performed outside the arena in a public space.
  - The robot starts at a predefined location.
- **People**: The operator is standing in front of the robot.
- **Objects**:
  - At least two bags are placed near the car.
  - A small object on the ground.
  - A hard-to-see object.
  - Retractable barriers.
- **Uncontrolled environment:** There are no restrictions on other people walking by or standing around throughout the complete task.

Procedure

1. **Following the operator:** The robot should inform the operator when it is ready to follow them. The operator walks naturally towards the car; after reaching the car, the operator points at the bag.
2. **Obstacles:** The robot will face 4 obstacles along its way (in arbitrary order): (a) a small object on the ground, (b) a hard-to-see object, (c) a crowd of people obstructing the path outside, and (d) a small area blocked using retractable barriers.
3. **Picking up the bag:** The robot picks up the bag pointed at by the operator.
4. **Bag delivery:** The robot takes the bag to the starting location. It may place the bag on the floor.
5. **Optional goals:**

   5.1. **Following the queue:** After the robot has reached the car, a few of the people that formed the crowd obstructing the robot return to the starting location in a queue. The robot can decide to join the queue on its way back, in a manner that appears natural to the people in the queue.

Additional Rules and Remarks

1. **Car location:** There is no real car outside; instead, a fixed location outside the *Arena* is designated as a car location.
2. **Reaching the car:** The robot can reach the car location only by following the operator (the location is unknown before the test).
3. **Deus ex Machina:** Score reductions for human assistance are applied in case the robot loses the operator, and needs to find them again through:

   3.1. Natural interaction (e.g., waving and calling)
   3.2. Unnatural interaction (e.g., raising both hands and jumping)
   3.3. Coming back (e.g., asking the operator to step back in front of the robot)
   3.4. Touching the robot (e.g., pulling the robot's hand)

   Note: Penalties for human assistance are only applied if the robot continues to score after the interaction. Penalties can be applied multiple times up to a total of 250 points.

Referee Instructions

The referees need to:

- Select one volunteer to act as the operator.
- Select three to four people to obstruct the robot's path outside and form the queue on the way back.
- Choose positions for the bags and assign a bag to the operator.
- Choose the order of the obstacles that the robot will face outside while following the operator.
- Designate a location outside as a car location.
- Designate a location for the queue to form. The queue is composed of the same people that form the crowd.
- Mind the robot when it moves around.
- Instruct the Operator to follow the robots instructions, step over the small obstacle and closely around the hard to see object

OC Instructions

At least two hours before test:

- Select and announce the robot's starting point.
- Select which bags will be used in the test.
- Organize retractable barriers.

Score Sheet

The maximum time for this test is **5:00** minutes.

Action	Score
Main Goal	
Following the person to the car	*300*
Perceiving the correct beg (visualize on screen or say which one)	*15*
Picking up the correct bag	*100*
Avoiding the crowd of people obstructing the path	*50*
Avoiding the small object on the ground	*50*
Avoiding the hard-to-see object	*50*
Avoiding the area blocked with retractable barriers	*50*
Bonus rewards	
Re-reaching the starting location	*200*
Joining and staying in the queue on the way to the arena	*300*
Penalties	
Dropping the bag	*–50*
Deus Ex Machina Penalties	
Rediscovering the operator by natural interaction	*–50*
Rediscovering the operator by unnatural interaction	*–100*
Rediscovering the operator by asking them to come back	*–150*
Rediscovering the operator by direct contact	*–200*
Special Penalties & Bonuses	
Not attending (see sec. 3.9.1)	*–500*
Using alternative start signal (see sec. 3.7.8)	*–100*
Outstanding performance (see sec. 3.9.3)	*111*

Total Score (excluding special penalties & standard bonuses) *1115*

5.2 General Purpose Service Robot

The robot is asked to understand and execute commands requiring a wide range of different abilities.

Main Goal: Execute 3 commands requested by the operator.

Optional Goals:

1. Understand a command given by a non-expert operator.

Focus

Task planning, *object/people detection and recognition*, *object feature recognition*, *object manipulation*

Setup

- **Locations:**
  - **Task location:** The task takes place inside the *Arena*, but some commands may require the robot to go out. The *Arena* is in its nominal state for this task.
  - **Start location:** The robot starts outside the *Arena*. When the door opens, the robot moves towards the *Instruction Point*.
  - **Instruction point:** At the beginning of the test, as well as after finishing the first and second command, the robot moves to the *Instruction Point*.
- **People:**
  - **Operators:** A *Professional Operator* (the referee) commands the robot to execute a task. Optionally, commands can be issued by a *Non-Expert Operator*, i.e. a person from the audience with no robotics background. In this case, the referee gives the goal of the command to the non-expert operator, who will then issue it to the robot in their own words. For example, the generated command might be "Bring me a coke from the kitchen." then the non-expert operator will be told "The robot should bring you a coke, which is found in the kitchen.", who then tells the robot "I want a coke. Go to the kitchen and get me one." If the robot consistently fails to understand the non-expert operator (e.g. after two retries), teams can default to a custom operator.

Procedure

1. **Instruction point:** At least two hours before the test, the referees announce the location of the *Instruction Point*.
2. **Test start:** The robot moves to the *Instruction Point* when the arena door is open.
3. **Command execution:** The operator instructs the robot to execute a command and the robot performs the task.
4. **Back to the instruction point:** The robot goes back to the *Instruction Point* and waits for the next command.

5. **Pausing the Timer:** The referee might pause the timer as soon as the robot reaches the instruction point to give time to setup the arena for the next command. The timer resumes as soon as the referee steps back in front of the robot for the next command.

Additional Rules and Remarks

1. **Partial scoring:** The main task allows partial scoring (per *completed* command).
2. **Command generator:** Tasks will be generated using the official command generator[1].
3. **Non-expert operators:** Referees are not allowed to instruct non-expert operators on how to operate the robot. Teams attempting to instruct or bias the operator will be disqualified from the task.
4. **Deus ex Machina:** The scores are reduced if human assistance is received, in particular for:

    - using a custom operator
    - bypassing speech recognition by using an alternative HRI
    - receiving human assistance to accomplish a task (score reduction is applied consistent with the other tasks in this rulebook)
    - instructing a human assistant to perform the whole task

Referee Instructions

- Provide the commands to the operators.

OC Instructions

At least two hours before the test:

- Generate the robot commands (don't reveal them to the teams!).
- Announce the location of the instruction point.
- Recruit volunteers to assist during the test.

During the test:

- Rearrange the arena so that it is in its nominal condition before each command.

Score Sheet

The maximum time for this test is **5:00** minutes.

[1] https://github.com/RoboCupAtHome/CommandGenerator

Action	Score
Main Goal	
Executing the task associated with each command	*3×400*
Bonus Rewards	
Understanding a command given by a non-expert operator	*3×100*
Deus Ex Machina Penalties	
Using a custom operator	*3×–50*
Bypassing speech recognition	*3×–50*
Instructing a human to perform parts of the task will apply a percentage penalty according to similar penalties in other Stage I tests.	*3×–400*
Special Penalties & Bonuses	
Not attending (see sec. 3.9.1)	*–500*
Using alternative start signal (see sec. 3.7.8)	*–100*
Outstanding performance (see sec. 3.9.3)	*150*

Total Score (excluding special penalties & standard bonuses) *1500*

5.3 Receptionist

Description

The robot has to take two new guests to the living room to introduce them and offer a free place to sit.

Main goal: The robot welcomes and assists two newcomers at a party, offering them a seat and maintaining appropriate gaze direction during conversation (at person speaking, direction of navigation).

Optional goals:

1. Open the entrance door for each arriving guest.
2. Visually describe the first guest for the second guest before reaching the living room
3. Identify similarities between the guests and the host and incorporate them into the conversation.

Focus

System Integration, *Human-Robot Interaction*, *Person Detection*, *Person Recognition*

Setup

- **Location**:
  - The test takes place in the living room.
  - The robot starts inside the *Arena* at a predefined location.
  - A table with drinks is prepared near the living room
  - **Entrance:** The entrance door is open by default. The team leader can request to close the door to score additional points by opening it for the guests.
- **People:**
  - **Host:** The host's name, favorite drink and interest will be announced before the test. The host is the only person sitting in the living room until the robot offers a free seat to another guest.
  - **Arriving Guests:** Both guests have a name, favorite drink, and a interest. An arriving guest will either step in front of the robot or wait behind the door to step in if the door is closed. Guests have to be guided to the beverage area and then the living room where the robot will introduce the guests to each other. Each of the guest will arrive separately.
  - **Passive Guests:** Other Guests: A few individuals, including possibly spectators, are standing in small groups of two or three, all over the arena but not covering relevant places e.g. the couch, seats or the area with the drinks.

Procedure

Both guests arrive separately. The robot either opens the door for the guest or waits for them at the starting point. It greets the guest and asks for their name. The robot then guides the guest to the beverage area, where it asks for their favorite drink and checks if the drink is available on

the table and where it stands. The robot finds out the interest of the guest at a freely chosen moment. After showing the guest the beverage area, the robot escorts them to the living room and offers a free seat. Once both guests are seated, the robot introduces them to each other.

- **Greeting guests:** The person paces a little to the left and right during conversation with the robot. Other people might appear in the background
- **Looking at person:** During verbal interactions and descriptions of people, robot looks at the conversational partner. The conversational partner will make small movements to each side to confirm the robot is dynamically looking at the person. Points for looking at the person talking, will only be awarded if the robot proves to continuously look at the moving person.
- **Looking at direction of navigation:** During navigation robot looks in the direction where it is going. Persistently gazing towards unrelated person or incorrect direction while moving during the task deducts points.
- **Smalltalk** Ask each guest for one interest.
- **Finding the drink** The robot shows both guest the drinking area where it will ask for their favorite drink and tell the guest if and where (left, center right is enough) that drink is available on the table.
- **Seating People:** The robot must point at a place or location where the guest can sit.
- **Switching Places:** Guests may switch places after they were seated.
- **Introductions:** When introducing guests, the robot must clearly identify the person being introduced and state their name favorite drink and an interest. Introducing two people mean to introduce them to each other.

Additional rules and remarks

1. **Opening Door Timing:** The Time of the test only starts after the fist person enters the arena or 2mins after start signal.
2. **Misunderstanding:** Not understanding the guests and asking them again is fine. Continuing with a wrong name, drink or interest causes a score reduction of 20pts per item.
3. **Partial Scoring:** The main task allows partial (per guest) scoring.
4. **Additional Seating:** Referees may add chairs to the party area living room.
5. **Deus ex Machina:** Score reduction applies per guest as follows:
    - **Custom Operator:** Since the main focus of the test is HRI, no custom operator can be chosen.
    - **Alternative HRI:** Using an alternative HRI to understand a guest causes a score reduction of 75pts.
    - **Recognizing People:** If the robot has to ask for help to identify people, score is reduced by 200pts.

Instructions

To Referee

The referees need to:

- Assign name, drink and one interest to 3 volunteers.

- Arrange (and re-arrange) people in the living room.
- Change the selection in the beverage area.
- Open the door when requested by the robot.

To OC

During setup day:

- Announce beverage location.

At least two hours before test:

- Announce starting position.
- Announce host's name, favorite drink and interest.
- Recruit five volunteers: one volunteer as host, two as arriving guests and 2 as passive guests.

Score sheet

The maximum time for this test is **6:00** minutes.

Action	Score
Main Goal	
Offer a free seat to the new guest	*2×100*
Show the guest around (navigate to the beverage area and living room)	*2×30*
Look in the direction of navigation or at the navigation goal	*2×15*
Tell position of favorite drink	*2×20*
Look at the person talking	*2×75*
Introduce both guests to each other	*180*
Bonus Rewards	
Open the entrance door for a guest	*2×200*
State a similarity between an interest between two or more persons	*50*
Describe the first guest to the second guest before reaching the living room (per correct visual attribute)	*4×30*
Penalties	
Wrong guest information was memorized (continue with wrong name, drink or interest)	*6×–20*
Describe the first guest to the second guest before reaching the living room (per incorrect visual attribute)	*4×–30*
Introduce the wrong persons	*–120*
Deus Ex Machina	
Alternative HRI	*2×–75*
Not recognizing people	*2×–200*
Special Penalties & Bonuses	
Not attending (see sec. 3.9.1)	*–500*
Outstanding performance (see sec. 3.9.3)	*123*
Total Score (excluding special penalties & standard bonuses)	*1230*

5.4 Storing Groceries

The robot stores groceries into a cabinet with shelves. Objects are sorted on the shelves based on similarity, for instance an apple is stored next to other fruits.

Main goal: Move objects from a table to the cabinet, grouping them by category or similarity. Refilling the cereal container.

Optional goals:

1. Opening the cabinet doors
2. Moving a *tiny* object
3. Moving a *heavy* object
4. Picking Objects from the shopping bag

Focus

Object detection and recognition, *object feature recognition*, *object manipulation.*

Setup

- **Locations:**
  - **Start location:** Before the test, the robot waits outside the *Arena* and navigates to the testing area when the door is open.
  - **Test location:** The testing area has a cabinet and a table nearby.
- **People:**
  - No people are involved in the test, unless the robot requires human assistance.
- **Furniture:**
  - **Table:** The table has 5–10 objects placed on it and the robot can choose which ones to grasp and in what order. On small tables, objects will be added as the robot frees up space.
  - **Shopping Bag:** There is a shopping bag with 5–10 additional normal items on the ground next to the table.
  - **Cabinet:** The cabinet contains objects arranged in groups — either by category or likeliness — on different shelves.
  - **Cabinet door:** The cabinet is closed. If the robot fails to open the door, it must clearly state this and request the referee to open it.
- **Objects**:
  - **Table objects:** The object on the table are arranged arbitrarily.
  - **Cabinet objects:** The objects are placed behind the cabinet doors and cannot be accessed unless the doors are open.
  - **Containers:** The container for the cornflakes are placed on the table.

Procedure

1. **Table location:** At least two hours before the test, the referees announce the table and cabinet that will be used in the test, as well as a rough location of the table.
2. **Test start:** The robot moves to the testing area when the arena door is open.
3. **Storing groceries:** After identifying the table, the robot moves the objects from the table to the cabinet.
4. **Pouring cereal:** The robot should pour cereal into the designated open container.

Additional rules and remarks

1. **Table:** The table's rough location will be announced beforehand, having its position to the left, right, or behind the robot.
2. **Incorrect categorization:** The score is reduced if an object is stored on the cabinet, but not on a shelf with similar objects; this reduction is applied per incorrectly stored object.
3. **New category:** Objects that do not semantically belong to any of the categories represented on the shelves should be grouped together on a new shelf.
4. **Deus Ex Machina:** The scores are reduced if human assistance is received, in particular for:
   - telling or pointing out to the robot where to place an object
   - moving an object instead of the robot
   - opening the cabinet doors
5. **Communicating Perception**: The robot must clearly communicate its perception to the referee. Pointing at the object or attempting to pick it up is sufficient. When using visualization, make sure the robot tells the referee where to look and make the visualization easily accessible. If the team wants to utilize bounding boxes make sure **only** one object with a bounding box is shown at a time, so the referee is easily able to check and verify. Also make sure the surrounding scene is visible, i.e. just showing a cropped bounding box is not enough.

OC Instructions

At least two hours before the test:

- Announce which table and cabinet will be used in the test.
- Announce a rough location for the table.
- Select which bags will be used in the test.
- Select which container will be used in the test.

Referee Instructions

The referee needs to:

- Place 5–10 objects on the table.
- Place container and cornflakes on the table.
- Place 5–10 objects in a bag near the table.
- Place objects in the cabinet, grouping them by category or likeliness.
- Close the door of the cabinet.

Score sheet

The maximum time for this test is **7:00** minutes.

Action	Score
Main Goal	
Navigating to the table	*15*
Perceiving object and categorizing it correctly	*5×15*
Picking up an object for transportation to the cabinet	*5×50*
Perceiving objects in shelf and saying on which layer the currently handled object should be placed	*5×15*
Placing an object in the cabinet	*5×15*
Placing an object next to similar objects on the cabinet	*5×50*
Opening the first cabinet door	*200*
Opening the second cabinet door	*100*
Pouring cereal into the container	*300*
Bonus Rewards	
Picking up an object from the shopping bag	*5×50*
Picking up a tiny object	*70*
Placing a tiny object	*30*
Picking up a heavy object	*70*
Placing a heavy object	*30*
Autonomously Picking any Object	*50*
Autonomously Placing any Object	*50*
Deus Ex Machina Penalties	
Perceiving object and categorizing it wrongly	*10×–15*
A human handing an object over to the robot	*5×–50*
A human placing an object in the cabinet	*5×–15*
A human placing an object in the cabinet next to similar objects	*5×–50*
A human pointing at a target location	*5×–25*
A human opening the cabinet door	*2×–100*
Spilling cereal while pouring	*–100*
Leaving cereal in the box	*–100*
A human pouring cereal in the bowl	*–300*
Special Penalties & Bonuses	
Not attending (see sec. 3.9.1)	*–500*
Using alternative start signal (see sec. 3.7.8)	*–100*
Outstanding performance (see sec. 3.9.3)	*189*
Total Score (excluding special penalties & standard bonuses)	*1890*

Chapter 6

Tests in Stage II

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6.1 Clean the Table

The robot cleans up a table after a meal. All dishes and cutlery must be placed inside the dishwasher and drinks must be placed inside the trash.

Main goals: Clean up all the objects on the table. Tableware and cutlery have to be placed correctly inside the dishwasher and drinks must be placed inside the trash bin.

Optional goals:

1. Opening the dishwasher door
2. Pulling out the dishwasher rack
3. Placing a dishwasher tab inside the dishwasher
4. Wipe the table area where the drink was

Focus

Object perception, *manipulation in narrow spaces*, and *task planning*.

Setup

- **Locations:**
  - **Start Location:** Before the test, the robot waits outside the Arena and navigates to the testing area when the door is open.
  - **Test location:** This test takes place in the kitchen.
- **People:**
  - No people are involved in the test, unless the robot requires human assistance.
- **Furniture:**
  - **Dishwasher:** A dishwasher is located close to the dining table.
  - **Tray:** A plastic tray, which may have other tableware and cutlery placed inside, is located either on top of the dishwasher or on one of the racks. Objects can be placed either in the dishwasher rack or in the tray, based on the team's choice.
  - **Trash bin:** A trash bin is located in the kitchen.
- **Objects:**
  - **Table setting:** The table has a total of seven objects disposed in a typical setting for a meal for one person. The cutlery and tableware objects may be randomly stacked as it is common after a meal. The object distribution is as follows:
    - *Cutlery*: Any two objects (fork, knife, or spoon).
    - *Tableware*: Any three objects (except cutlery), at least one of which is a dish.
    - *Drinks*: Any two drinks from the object set.
  - **Dishwasher tab:** The tab can be found at a location announced before the test and should be autonomously placed inside the tab slot in the dishwasher.
  - **Wiping Object:** The wiping object (i.e. a sponge) can be found at a location announced before the test.

Procedure

1. **Dishwasher door and rack**: Half an hour before the test, the team informs the referees:
   - whether objects will be placed in the dishwasher rack or in the tray
   - whether the dishwasher door should be closed and, if the door is open, whether the rack should be pushed in
2. **Test start:** The robot moves to the kitchen when the arena door is open.
3. **Table clean up:** The robot cleans the table by putting: the cutlery and tablware items inside the dishwasher and the drinks in the trash bin.

Additional Rules and Remarks

1. **Safe placing:** Objects must be placed with care, namely the robot should place rather than throw or drop objects.
2. **Dishwasher:** The team decides whether the robot will place the objects in the dishwasher's rack or in the plastic tray.
3. **Dishwasher door:** The dishwasher door is open and with the racks pulled out by default. The team leader can, however, request the dishwasher to be closed for bonus points. If the robot fails to open the door, it must clearly state this and request the referee to open the door.
4. **Correct item placement:** The items should be placed in the rack/tray correctly, namely in the location and in a manner as a person would put them.
5. **Deus ex Machina:** The scores are reduced if human assistance is received, in particular:
   - pointing to an object or telling the robot where an object is
   - handing an object over to the robot
6. **Communicating Perception**: The robot must clearly communicate its perception to the referee. Pointing at the object or attempting to pick it up is sufficent. When using visualization, make sure the robot tells the referee where to look and make the visualization easily accesible. If the team wants to utilize bounding boxes make sure **only** one object with a bounding box is shown at a time so the referee is easily able to check and verify. Also make sure the surrounding scene is visible, i.e. just showing a cropped bounding box is not enough.

OC Instructions

During the *Setup Days*:

- Provide official cutlery and tableware for training.
- Announce what is the object used for wiping.

At least two hours before the test:

- Announce the predefined location of the dishwasher tab.
- Announce the predefined location of the object used for wiping.

Referee Instructions

The referee needs to:

- Place objects on the table (two pieces of cutlery, three pieces of tableware and two drinks)
- Place the plastic tray on the dishwasher or on a rack, as requested by the team.

Score sheet

The maximum time for this test is **10:00** minutes.

Action	Score
Main Goal	
Navigate to the table to pick up items	*15*
Picking up the cup, bowl and drinks for transportation	*4×50*
Picking up cutlery (spoon, fork) for transportation	*2×80*
Picking up the plate for transportation	*100*
Placing the tableware and cutlery inside the dishwasher	*5×50*
Placing an item correctly (cleanable, convenient like a human would) in the dishwasher	*5×75*
Placing a drink inside the trash bin	*2×50*
Bonus Rewards	
Pulling and pushing the dishwasher rack	*2×100*
Opening and closing the dishwasher door	*2×200*
Picking up the dishwasher tab for transportation to the dishwasher	*100*
Placing the dishwasher tab inside the dishwasher's hatch intended for the tab	*200*
Wiping the area where the drink was	*2×50*
Autonomously Picking any Object	*50*
Autonomously Placing any Object	*50*
Deus Ex Machina Penalties	
Handing cup, bowl and drinks over to the robot	*3×–50*
Handing cutlery over to the robot	*2×–80*
Handing the plate over to the robot	*–100*
Having a human place an object in the dishwasher	*5×–50*
Having a human place a drink inside the trash bin	*2×–50*
A human pointing at the trash bin	*2×–25*
Special Penalties & Bonuses	
Not attending (see sec. 3.9.1)	*–500*
Using alternative start signal (see sec. 3.7.8)	*–100*
Outstanding performance (see sec. 3.9.3)	*230*

Total Score (excluding special penalties & standard bonuses) *2300*

6.2 Enhanced General Purpose Service Robot

The robot is asked to maintain the household by cleaning up the arena and assisting people.
Main Goal: Solve different problems in the arena.

Focus

Task planning, object/people detection and recognition, object feature recognition, object manipulation

Setup

The arena is in its default state apart from problems set up for the robot to solve:

1. **Problems:**
   - **Trash:** Objects on the floor are to be thrown in the trash.
   - **Objects:** Objects that are not in their default location should be returned there (see 3.3.5).
   - **Persons:** Some persons in the arena will have requests for the robot. They will raise their hand if the robot is in the same room.

Procedure

1. **Test start:** The robot enters when the arena door is open.
2. **Finding Problems:** The robot has to find problems to solve on its own.

Additional rules and remarks

1. **Number of Problems:** The number of problems depends on the arena size the minimum count of generated problems is 8.
2. **Repeating Problem Category:** Solving the same Category of Problem incurs a penalty.
3. **Solving more:** You can continue solving problems to compensate for penalties.
4. **Partial Scoring:** The main task allows partial scoring (per *solved* problem).
   4.1. **Scores:** Score reduction is applied as a percentage depending on the solution.
   4.2. **Penalties:** The Repetition penalty is applied before any partial penalties
   4.3. **Example:** If Picking up trash off the floor is valued as 60% of the solution. Requesting handover should be a $650 \times 0.6 = 390$ points penalty for the first pick and $(650 - 300) \times 0.6 = 210$ for the second pick.
5. **Command Generator:** Problems and commands will be generated using the official command generator[1].
6. **Finding People:** Finding a person and stating they need help counts as finding the problem.
7. **Understanding Commands:** Understanding and correctly repeating the command given by a person can be counted towards partially solving a problem.

[1] https://github.com/RoboCupAtHome/CommandGenerator

Referee instructions

- Setup the problems in the arena.
- Provide commands to volunteers.

OC instructions

At least two hours before the test:

- Generate the problems and commands (don't reveal them to the teams!).
- Recruit volunteers to assist during the test.

Score sheet

The maximum time for this test is **10:00** minutes.

Action	Score
Main Goal (can be repeated unlimited times)	
Find and clearly state an encountered problem	*3×150*
Solve a problem	*3×650*
Penalties	
Find repeated problem category	*–100*
Solving repeated problem category for the 2nd time	*–300*
Solving repeated problem category for the 3rd (or more) time	*–500*
These Penalties are applied before any percentage based Deus Ex Machina Penalties	
Deus Ex Machina Penalties	
Asking for location of a problem	*3×–150*
Instructing a human to perform parts of the task will apply a percentage penalty according to similar penalties in other Stage II tests.	*3×–650*
Special Penalties & Bonuses	
Not attending (see sec. 3.9.1)	*–500*
Using alternative start signal (see sec. 3.7.8)	*–100*
Outstanding performance (see sec. 3.9.3)	*240*

Total Score (excluding special penalties & standard bonuses) *2400*

6.3 Restaurant

Description

The robot retrieves and serves orders to several customers in a real restaurant previously unknown to the robot.

Main goal: Detect calling or waving customer, reach a customer's table without prior guidance/training. Take and serve all orders.

Optional goal: Use an unattached tray to transport the order.

Focus

This task focuses on *Task planning*, *Online mapping*, *Navigation in unknown environments*, *Gesture detection*, *Verbal interaction* and *Object manipulation*.

Setup

- **Locations:**
  - This task takes place in a real restaurant fully equipped and in business. When this is not possible, the test can be conducted in any place with the appropriate locations other than the *Arena*.
  - The *Restaurant* location will remain secret until the start of the test.
  - The robot starts next to the *Kitchen-bar*. It is a table located near the restaurant's kitchen.
- **People:**
  - A *Professional Barman* (member of the TC) awaits at the other side of the *Kitchen-bar* for orders to be placed. The *Professional Barman* assists the robot on request.
  - There may be real customers and waiters around.
  - There are at least three tables occupied with professional customers (member of the OC/TC).
  - There are at least two tables occupied with regular customers.
  - Customers may call the robot any time, even simultaneously.
- **Furniture:** The furniture is not standardized and will be kept the same as the restaurant or place selected for the task.
- **Objects:**
  - Objects to fulfill orders are located on the Kitchen-bar.
  - Orders have two or three objects randomly chosen.
  - All edible/drinkable objects from the list of standard objects (see Section 3.3.5) are eligible to be part of the orders.

Procedure

1. The referee requests the team to move the robot to the start location.
2. The referee gives the start signal and starts the timer.
3. The team leaves the area after the start signal.

4. A TC member follows the robot ready to press the emergency stop button.
5. The robot detect calling or waving customer and reach a customer's table.
6. The robot take the customer's order, place the order, and deliver it.
7. **Optionally**, the robot can use an unattached tray to transport the order.

Additional rules and remarks

- **Remarks:**
  - This test takes place in a public area. The robot is expected to not even slightly touch anyone or anything and is immediately stopped in case of danger.
  - Since this task is performed outside the arena, the time limit may be longer than the others tasks.
  - The availability of wireless, external computing devices, or electrical outlets can't be guaranteed. Assume unavailability.
  - The robot interacts with the operators, not the team. The team is not allowed to instruct anyone. All instructions should be provided by the robot itself.
  - The robot may use up to one minute to instruct the *Professional Barman.*
  - The robot can request to be guided to a customer's table.
  - The robot can choose to take several orders and place them later on, place an order and pick the next one while the former is being served, or dispatch one order at a time.
  - The robot should politely confirm the order to the client when receiving it, keeping the guest pleased.
  - The robot can either transport each object individually, or using a tray. All delivered objects must be placed on the customer's table.
  - For transport with an unattached tray, the robot must pick up the objects and place them on a tray, pick up the tray, and then place the objects from the tray on the table (first placing the tray on the table is allowed).
  - If requested, the barman will place the order in a basket or tray for the robot to deliver it.
  - Upon arrival to the restaurant, only two team members are allowed next to the robot for watching and charging.
  - If a person from the audience (severely) interferes with the robot in a way that makes it impossible to solve the task, the teams may repeat the test immediately.
  - Each Deus Ex Machina penalty for skipping manipulation will only be applied twice per order so receiving an order with three objects is not more punishing.
  - If the robot detects a customer but does not reach their table, the robot must clearly show who was detected to receive points, i.e. displaying a picture of the person.
  - When a team is at the front of the queue, they are allowed to begin their startup procedure (the robot must remain in place). When it is their turn, they must bring the robot directly and in a straight line from the front of the queue to the start location. Once at the start location only slight movements are allowed (no moving back and forth, no full rotations etc.)
- **Disqualification:**
  - Touching the robot after the start signal.

– Mapping the area in advance.

Instructions:

To Referee

The referee needs to:

- Prepare orders for each client.

To OC

The OC needs to:

- **During Setup days:** Check with local (security) management if the possible location, including a sufficient queuing area, can be used for the restaurant test.
- **1 hour before the test:** Gather all teams and robots to move to some nearby queuing area and instruct the teams how/when to move to the actual test location.

Score sheet

The maximum time for this test is **15:00** minutes.

Action	Score
Regular Rewards	
Detect calling or waving customer	*2×100*
Reach a customer's table without prior guidance/training	*2×100*
Understand and confirm the order received to the customer	*2×200*
Communicate the order to the barman	*2×100*
Return to the customer table with the order	*2×100*
Serve the order to the customer	*2×200*
Bonus Rewards	
Use an unattached tray to transport	*2×200*
Regular Penalties	
Being guided to a table	*2×–200*
Not making eye-contact when taking an order	*2×–80*
Not reaching the bar (barman has to move from behind the bar to interact with the robot)	*2×–80*
Deus ex Machina Penalties	
Asking the Barman to handover object to the robot	*4×–50*
Guest needing to take the object from a tray or the robot's hand	*4×–50*
Being told/pointed where is a table/*Kitchen-bar*	*2×–100*
Special Penalties & Bonuses	
Not attending (see sec. 3.9.1)	*–500*
Using alternative start signal (see sec. 3.7.8)	*–100*
Outstanding performance (see sec. 3.9.3)	*200*

Total Score (excluding special penalties & standard bonuses) *2000*

6.4 Give Me a Hand

Description

The robot places new items in the locations given by a human through natural human-robot interaction.

Main goal: New items have arrived at the house. The user takes each item from the box or bag, one by one, hands it to the robot, and instructs the robot to place it in a specified location.

Focus

This task focuses on *Object perception*, *Human perception*, *Verbal interaction*, and *Human-Robot Interaction*.

Setup

- **Locations:**
  - This test takes place inside the Arena.
- **People:**
  - There might be several people inside the room (e.g. referee, assistant, and the operator).
- **Objects:**
  - There is a box or bag with five random known objects (see 3.3.5).
  - The arena must include visually similar delivery locations (e.g., identical bins or dishes) to create deliberate ambiguity requiring user clarification.

Procedure

1. The robot starts outside the arena and navigates to the testing area when the door is open.
2. The robot approaches the calling operator. The operator's location remains the same for the duration of the test.
3. The operator takes an item out of the box and hands it to the robot.
4. The operator instructs the robot to place the object in a specified location in the same room.
5. The robot delivers the item and returns to the initial operator's location for a new object.

Additional rules and remarks

1. **Handover:**
   - To ensure user safety, the robot must position its manipulator within 10-20 cm of the human's hand, without making physical contact, and must clearly indicate its readiness to receive the object using an explicit signal (e.g., holding position, activating a light, saying it, or playing a sound).
   - The human must complete the handover by placing the object into the robot's manipulator; the robot must remain stationary and must not initiate physical contact during this phase.
   - Explicit operator commands during the object handover—such as *"Please, place the item in my gripper and wait until I close it in three, two, one"*—will be considered *Deus ex Machina*. However, the robot may notify the user before closing its gripper after detecting that the object is within its grasping range; for example, *"I have detected that the object is in my hand. I will close my gripper in three seconds—please be careful."*
2. **Delivery:**
   - Delivery instructions may include phrases like: *"Place this apple on that plate"* or *"Put this can in the trash bin."* The operator gives instructions using both verbal commands and gestures (e.g., *"Place this item there (pointing), next to the flower vase"*).

- At any moment, the robot may ask questions to confirm information, request clarification, or ask the operator to repeat the instruction.
- Since the target locations are intentionally ambiguous, the robot can request clarification from the operator before proceeding. This should be done through natural interaction, such as verbal questions or multimodal cues (e.g., pointing, asking "Which one?").
- Full points for a correct delivery will only be awarded if the object is placed inside the designated container (e.g., bins) or within a few centimeters of the intended location (e.g., plates or surfaces).
- In this task, natural human-robot interaction is the primary focus. Before engaging with the operator to request clarification, the robot must first capture their attention — for example, through visual signals, gestures, or sound cues — rather than relying on unattended interactions (e.g., talking without addressing a specific person). This is particularly important when the operator and the robot are not in close proximity; for instance, the operator might be occupied retrieving the next object or speaking with the referee.

Instructions:

Referee instructions

The referee needs to:

- Select random objects from the dataset and target locations.
- Assing the operator's location in the room.

OC Instructions

At least two hours before the test:

- Announce the room for the test.
- Arrange target containers and placement items (such as similar bowls, dishes, or bins) in the delivery areas.

Score sheet

The maximum time for this test is **10:00** minutes.

Action	Score
Regular Rewards	
Approach a calling operator	*100*
Approach the human's hand without contact	*5×50*
Receive an object from the operator	*5×100*
Use natural interaction to clarify ambiguous delivery locations	*5×50*
Place an object at the correct location	*5×100*
Penalties	
Initiating unattended interaction (e.g., talking to the open air)	*5×−50*
Place an object at the wrong location or by guessing the intended location	*5×−75*
Deus Ex Machina Penalties	
Instruct the user how to hand over an object	*5×−75*
Special Penalties & Bonuses	
Not attending (see sec. 3.9.1)	*−500*
Using alternative start signal (see sec. 3.7.8)	*−100*
Outstanding performance (see sec. 3.9.3)	*160*

Total Score (excluding special penalties & standard bonuses) *1600*

Chapter 7

Finals

The competition ends with the Finals on the last day, where the two teams with the highest total score compete.

Even though each league has its own first, second and third place, the *Finals* are meant to show the best of all leagues to the jury members as well as the audience and, thus, warrants a single schedule slot.

7.1 Scoring

The final score and ranking are determined by the jury evaluations and by the previous performance (in Stages I and II) of the team, in the following manner:

1. The influence of the final ranking is 50 %.
2. The influence of the total sum of points scored by the team in Stage I and II is 50 %.

There is no maximum score during the Final.

These demonstrations are carried out in a serialized fashion, one League performing after another in one *Arena*.

7.1.1 Task

The procedure for the demonstration is the same as for Enhanced General Purpose Service Robot (see Task 6.2) with the added tasks of:

1. **Closing Furniture:** Doors of the Cabinet as well as the Dishwasher need to be closed.
2. **Welcome Guest:** There is an additional person waiting behind the exit door. The person will state their request after being welcomed by the robot. The door must be opened without human assistance. As the position is known, there will be no points awarded for finding this person.
3. **Custom Tasks:** Additional reasonable household task.

Custom Tasks

1. **Requesting Additional Tasks:** Additional reasonable household tasks may be requested for scoring during the team leader meetings. These tasks must be approved by the *technical committee* (TC).
2. **Arena-Specific Tasks:** Depending on the arena setup, certain household chores may be delegated to the robot for completion. Examples of tasks that may be handled by the robot include, but are not limited to, window cleaning and picture alignment on the wall.
3. **Team-Supplied Items** In the event that additional items are necessary for the completion of a task (e.g., clothing for folding, watering cans), the requesting team is responsible for supplying these items.
   3.1. **Standardization** These items must be regular household items (no markers, no custom printed handles etc.).

3.2. **Availability** All requested items must be provided and made available to the competitors prior to the day of the final. At least two identical copies of each item must be supplied to ensure adequate access.

7.1.2 The Show Must Go On

To ensure a good experience for the audience the teams are allowed to restart the robot inside the arena.

1. All previously collected points will be kept.
2. The tasks may be rearranged by the Referee during a restart.
3. The restart penalty is only applied if the robot continues scoring afterwards.

7.1.3 Commentator

The team will be asked to provide a commentator to explain the robot's behavior and answer questions to produce a better viewing experience for the audience.

Score sheet

The maximum time for this test is **10:00** minutes.

Action	Score
Main Goal (can be repeated unlimited times)	
Find and clearly state an encountered EGPSR problem	*3×150*
Solve a EGPSR problem	*3×650*
Opening the Door of the Apartment	*600*
Closing the Dishwasher	*600*
Closing a Cabinet Door	*300*
Custom Task	
Custom Task	
Custom Task	
Penalties	
Find repeated EGPSR problem category	*–100*
Solving repeated EGPSR problem category for the 2nd time	*–300*
Solving repeated EGPSR problem category for the 3rd (or more) time	*–500*
Deus Ex Machina Penalties	
Asking for location of a problem	*3×–150*
Restart (only applies if the robot continues scoring afterwards)	*–50*
Instructing a human to perform parts of the task will apply a percentage penalty according to similar penalties in other Stage II tests.	*3×–650*

Total Score (excluding special penalties & standard bonuses) *3900*

7.2 Final Ranking and Winner

The winner of the competition is the team that gets the highest ranking in the *Finals*.
The second place will be the team that got the second-highest ranking in the *Finals*.
The third place will be the team with the lowest score in the *Finals*.

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